Ansh Chandrakar

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LinkedIn Profile Portfolio Website

ABOUT ME

MSc Physics student at IIT Hyderabad with a strong foundation in computational physics. Passionate about exploring astrophysics and high-energy phenomena through numerical modeling and simulation. Experienced in utilizing Python to bridge theoretical physics with computational reality.

EDUCATION

Master of Science in Physics 2025 - Present
Indian Institute of Technology, Hyderabad, Telangana
- *CGPA*: 6.71/10

Bachelor of Science (Honours) in Physics 2022 - 2025
Birla Institute of Technology, Mesra, Ranchi, Jharkhand
- *CGPA*: 8.6/10
- *Elective Courses*: Computational Physics Skills, Radiation Safety, Nuclear and Particle Physics, Classical Dynamics, Astronomy and Astrophysics

High School 2017 - 2021
Sri Sankara Vidyalaya, Bhilai, Chhattisgarh
- *12th*: 91.8%
- *10th*: 92.4%

RESEARCH PROJECTS

Academic Coursework: SWIR Optical-CDMA Encryption Modeling
- Engineered Digital Signal Processing (DSP) algorithms in Python (NumPy) to cryptographically map data to specific wavelengths and amplitude weights for a hybrid physical-layer encryption system.
- Simulated a stochastic noise layer by generating 300 discrete Gaussian decoy peaks, enforcing strict spectral guard bands to obfuscate true data channels.
- Validated the computational protocol using Ansys Zemax OpticStudio, utilizing high-density geometric ray-tracing to model physical irradiance and spatial dispersion.

MSc Winter Project: Non-Reciprocal Ising Model with Stochastic Resetting
- Simulated the non-equilibrium statistical mechanics of Ising systems with asymmetric (non-reciprocal) interactions violating detailed balance.
- Developed numerical models to analyze stochastic spin-flip dynamics, tracking the emergence of non-equilibrium steady states and probability currents.
- Quantified the effects of stochastic resetting on system dynamics through computational analysis, highlighting the interplay between non-reciprocity and resetting.

Bachelor's Dissertation: Entanglement in Coupled Quantum Oscillators
- Programmed robust quantum simulations using the Quantum Toolbox in Python (QuTiP) to explore entanglement in coupled quantum harmonic oscillators.
- Numerically solved for unitary evolution and computationally derived entanglement measures, including Von Neumann entropy and logarithmic negativity, across various system states.

- Modeled hybrid systems coupling quantum oscillators with coherent states, generating computational visualizations to analyze entanglement dynamics and decoherence effects.

EXPERIENCE

Winter School on Recent Frontiers in Physics Research (Online) December 2025
Indian Institute of Science (IISc), Bengaluru

- Participated in advanced lectures covering astrophysics, cosmology, quantum condensed matter, and plasma physics.
- Completed intensive computational tutorials, directly applying numerical methods and programming to model complex physical phenomena discussed in lectures.
- Certificate of Participation

IIA Online Summer School June 2025
Indian Institute of Astrophysics, Bengaluru

- Selected to participate in a rigorous summer school hosted by the Indian Institute of Astrophysics.
- Attended lectures and hands-on data sessions on advanced topics including galactic structure, observational techniques, radio astronomy, and cosmology.
- Certificate of Completion

Society for Data Science (Member) Feb 2023 – May 2025
BIT, Mesra, Ranchi

- Upskilled in computational data analysis and numerical programming through intensive Python workshops.
- Spearheaded logistics and event coordination for society-wide technical coding initiatives and hackathons.
- Secured event sponsorships by pitching society technical initiatives to external industry partners.

CERTIFICATIONS & AWARDS

- **IIT JAM 2025**: Achieved All India Rank (AIR) 220 in the Joint Admission Test for Masters.
- **BIT GP Birla Scholarship for Program Topper**: Awarded twice (2 semesters) for outstanding academic performance at BIT Mesra